Section 9.1 Linear Transformations

Let V, W be vector spaces. Then $T: V \to W$ is a linear transformation if for every $\vec{v}_1, \vec{v}_2 \in V$

$(a)\ T(\vec{v}_1 + \vec{v}_2) = T(\vec{v}_1) + T(\vec{v}_2)$

This is pretty much the same definition we had in section 3.1, however now we allow for general vector spaces!

Let $T: V \to W$ be a a linear transformation $T(\vec{v}) = \vec{w}$.

We call the set of input vectors $\{\vec{v}\} \in V$ the **Domain** of T

We call the set of output vectors $\{\vec{w}\} \in W$ the **Codomain** of T

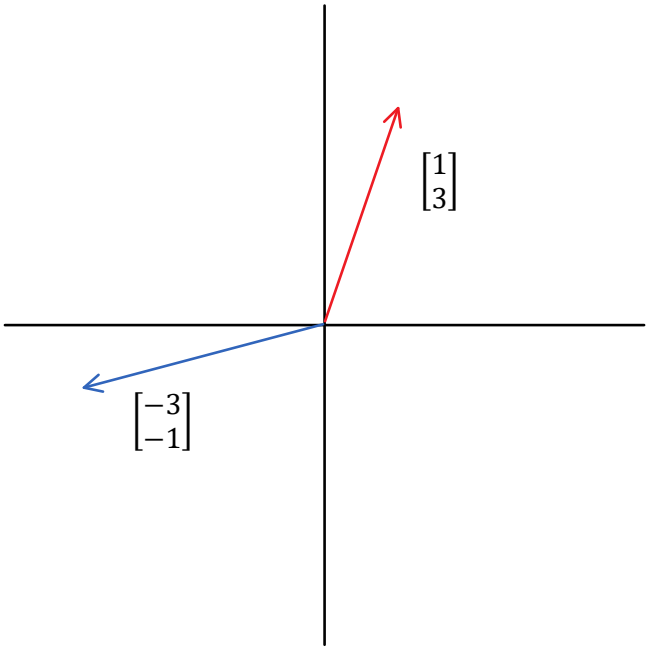
For a specific vector $\vec{v}$ in the domain of T we call $\vec{w} = T(\vec{v})$ the **Image** of $\vec{v}$ under T

The collection of the images of all of the vectors in the domain is called the **Range** of T (range(T)

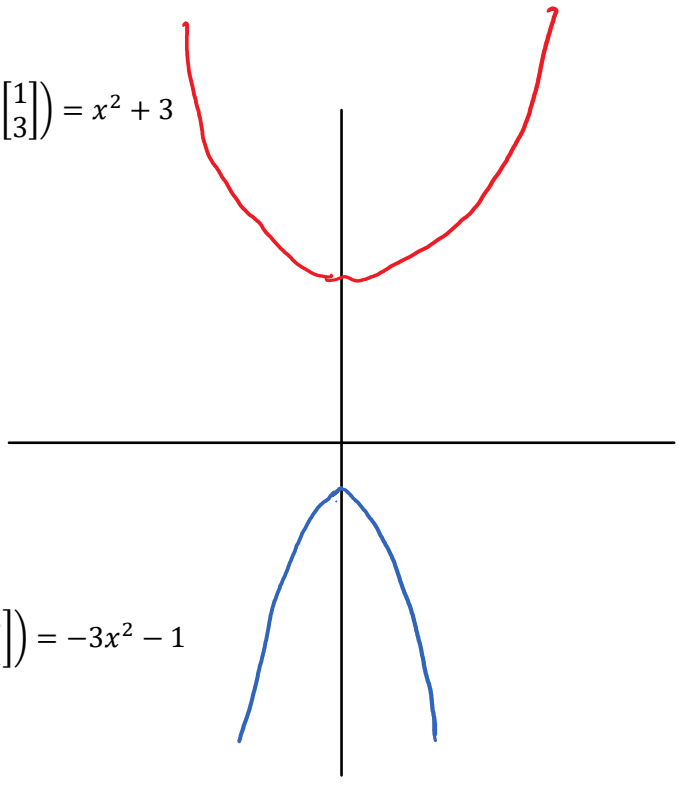
The set of all input vectors $\{\vec{v}\} \in$ V such that $T(\vec{v}) = 0_w$ is called the **Kernal** of T (ker(T))

Let $T: R^2 \to P^2$ be defined as

$$T\left(\begin{bmatrix} a \\ b \end{bmatrix}\right) = ax^2 + b$$



$$T\left(\begin{bmatrix} 1 \\ 3 \end{bmatrix}\right) = x^2 + 3$$



$$T\left(\begin{bmatrix} -3 \\ -1 \end{bmatrix}\right) = -3x^2 - 1$$

Section 9.1 Linear Transformations

Let $T: R^2 \to P^2$ be defined as $T\left(\begin{bmatrix} a \\ b \end{bmatrix}\right) = ax^2 + b$

Is it linear?

$(a)\ T(\vec{v}_1 + \vec{v}_2) = T(\vec{v}_1) + T(\vec{v}_2)$

$(b)\ T(c\vec{v}_1) = cT(\vec{v}_1)$

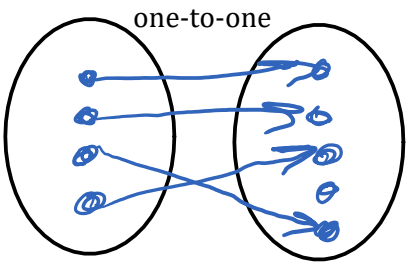
Let $T: V \to W$ be a linear transformation then ker(T) is a subspace of V and if S is a subspace of V then $T(s)$ is a subspace of W

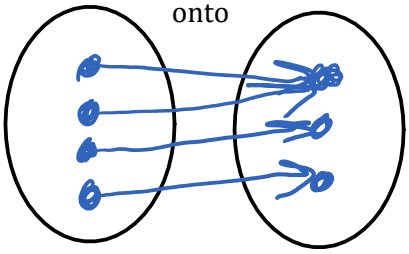
Proof:

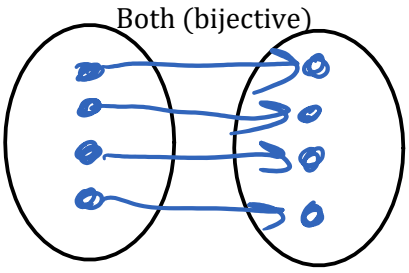
Section 9.1 Linear Transformations

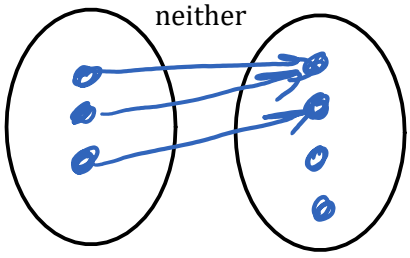
Let $T: V \to W$ be a linear transformation then

(a) T is one-to-one (injective) if for every vector $\vec{w} \in$ W there exists at most one vector $\vec{v} \in$ V such that $T(\vec{v}) = \vec{w}$.
(b) T is onto (surjective) if for every vector $\vec{w} \in W$ there exists at least one vector $\vec{v} \in$ V such that $T(\vec{v}) = \vec{w}$









Let $T: R^{2\times 2} \to R$ be defined as

$$T(A) = T\left(\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}\right) = tr(A) = a_{11} + a_{22}$$

Is T onto?

Is T $1-1$?

Section 9.1 Linear Transformations

Let $T: V \to W$ be a linear transformation, then T is 1-1 if and only if $T(\vec{v}) = O_W$ only has the trival sollution of $\vec{v} = 0_V$.

Suppose $T(\vec{v}) = O_W$ only hs the trival solution and that T is *not* 1-1. Then there should exist $\vec{v}_1, \vec{v}_2$ such that $T(\vec{v}_1) = T(\vec{v}_2) = \vec{w}$

$T(\vec{v}_1) - T(\vec{v}_2) = 0_w$ And because T is linear…

Suppose T is 1-1

Suppose there is some $\vec{v}_1 \neq 0$ such that $T(\vec{v}_1) = 0_w$

But $0 \cdot T(\vec{v}_1) = 0 \cdot 0_w$ and we proved in section 7.1 that….

Let V be a vector space and suppose that $\vec{v} \in V$. Then,

(a) $\vec{0}$ *is a vector in* V *then* $\vec{v} + \vec{0} = \vec{v}$
(b) *if* $-v$ *is the additive inverse of* $\vec{v}$ *then* $\vec{v} + (-\vec{v}) = \vec{0}$
(c) *the additive inverse of* $\vec{v}$ *is unique*
(d) $0 \cdot \vec{v} = \vec{0}$
(e) $-1 \cdot \vec{v} = -\vec{v}$

Section 9.1 Linear Transformations

Let $T: V \to W$ be a linear transformation. Suppose that $S_v = \{\vec{v}_1, \dots, \vec{v}_k\}$ is a subset of V and $S_w = \{\vec{w}_1, \dots, \vec{w}_k\}$ as subset of W with $T(\vec{v}_i) = \vec{w}_i$ for *each* vector in S_v. Then if S_w is linearly independent, so it S_v

Proof.

Start with $c_1\vec{v}_1 + c_2\vec{v}_2 + \cdots c_k\vec{v}_k = 0_V$ If the $\vec{v}'s$ are to be linearly independent than the only solution to the equation should be all of the $c's$ being zero.

Now we assume the $\vec{w}'s$ are linearly independent and we examine keeping in mind that T *is linear*

$$T(c_1\vec{v}_1 + c_2\vec{v}_2 + \cdots c_k\vec{v}_k) = T(O_V)$$

Prove that $\{\sin(x), \cos(x)\}$ on $C(R)$ are linearly independent by finding a linear transformation from them to the linearly independent vectors $\begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}0\\1\end{bmatrix}$ in R^2.

Section 9.1 Linear Transformations

Let $T: V \to W$ be a linear transformation with V and W finite dimensional. Then,

$$\dim(V) = \dim(\ker(T)) + \dim(range(T))$$

Proof:

Let $\{\vec{v}_1, \dots, \vec{v}_k\}$ be a basis for $\ker(T)$ so that $\dim(\ker(T)) = k$ now extend the list with $\{\vec{v}_{k+1}, \dots \vec{v}_m\}$ to obtain a basis for V of

$$\{\vec{v}_1, \dots, \vec{v}_k, \vec{v}_{k+1}, \dots \vec{v}_m\}$$

So here $\dim(V) = m$

Now take $\vec{v}$ in V using our basis it can be written in a unique way as,

$$\vec{v} = c_1\vec{v}_1 + c_2\vec{v}_2 + \cdots + c_k\vec{v}_k + c_{k+1}\vec{v}_{k+1} + \cdots + c_m v_m$$

Now we examine $T(\vec{v})$ keeping in mind that $\vec{v}_1 \dots \vec{v}_k$ are in $\ker(T)$. Our end goal will be to show that the linear transformation of all of the $\{\vec{v}_{k+1}, \dots \vec{v}_m\}$ form a basis for $range(T)$ if so then then it is $m - k$ dimensional and we wil have our theorem!

Section 9.1 Linear Transformations

Let $T: V \to W$ be alinear transformation with V *and* W finite dimensional then,

(a) If T is onto, then $\dim(V) \geq \dim(W)$

(b) If T is one-one then $\dim(V) \leq \dim(W)$

Proof:

If T is onto $w = range(T)$ so $\dim(W) = \dim(range(T))$

So $\dim(V) = \dim(\ker(T)) + \dim(W)$ so if there is any thing in the kernel we add that dimension to that of W and (a) is true.

If T is 1-1 then $\dim(\ker(T)) = 0$

So $\dim(V) = 0 + \dim(range(T))$ and either $range(T) = W$ or it is not onto and $range(T) \subset W$ and so b is true.